

Particle-scale CO₂ adsorption kinetics modeling considering three reaction mechanisms

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ABSTRACT

In the presence of water (H₂O), dry and wet adsorptions of carbon dioxide (CO₂) and physical adsorption of H₂O happen concurrently in a sorbent particle. The three reactions depend on each other and have a complicated, but important, effect on CO₂ capturing via a solid sorbent. In this study, transport phenomena in the sorbent were modeled, including the three reactions, and a numerical solving procedure for the model also was explained. The reaction variable distribution in the sorbent and their average values were calculated, and simulation results were compared with experimental data to validate the proposed model. Some differences, caused by thermodynamic parameters, were observed between them. However, the developed model reasonably simulated the adsorption behaviors of a sorbent.

The weight gained by each adsorbed species, CO₂ and H₂O, is difficult to determine experimentally. It is known that more CO₂ can be captured in the presence of water. Still, it is not yet known quantitatively how much more CO₂ the sorbent can capture, nor is it known how much dry and wet adsorptions separately account for CO₂ capture. This study addresses those questions by modeling CO₂ adsorption in a particle and simulating the adsorption process using the model. As adsorption temperature changed into several values, the adsorbed amount of each species was calculated. The captured CO₂ in the sorbent particle was compared quantitatively between dry and wet conditions. As the adsorption temperature decreased, wet adsorption increased. However, dry adsorption was reduced.

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1. Introduction

Anthropogenic emissions of carbon dioxide (CO₂), one of the most critical species in greenhouse gases (Pachauri and Reisinger, 2007), is considered as the main culprit to global climate change (Rogner et al., 2007). In the United States, about 83% of greenhouse gases are produced from the burning of fossil fuels (EIA, 2006a), which emits about 2 billion metric tons of CO₂ annually (EIA, 2006b) and amounts to roughly two-thirds of the total U.S. power sector CO₂ emissions (Ciferno et al., 2009). To mitigate the emission problem, various government agencies and industries have been developing carbon capture, transport, and long term sequestration technologies (Figuerola et al., 2008).

Generally speaking, three types of approaches are being examined for CO₂ capture from fossil-fuel-based power plants: (1) pre-combustion, (2) post-combustion, and (3) oxy-combustion. In pre-combustion capture, CO₂ is removed from the fuel stream before it is burned, and H₂ is sent to a combustion turbine. For example, a polymer-based membrane with high H₂/CO₂ selectivity was reported by Berchtold (2006). Supported ionic liquid

membranes (Ilconich et al., 2007) and lithium silicate-based sorbent (Li et al., 2006) were also developed to select CO₂ from coal-derived syngas. In oxy-combustion, fuel is burned with a mixture of nearly pure O₂ and recycled flue gas, and its flue gas contains mainly CO₂ and water. After removing water by condensation, highly concentrated CO₂ can be easily collected. The drawback of this approach is extracting O₂ from air, which is a costly process. Meanwhile, in post-combustion capture, CO₂ is captured from flue gas produced by the combustion process. Notably, existing power plants can be easily retrofitted with the equipment required for this process (Figuerola et al., 2008). Thus, it is considered the most feasible approach to implement in the near future.

As a post-combustion capture process, aqueous amine-based solvent capture is commercially available, but it is a highly energy-intensive and corrosive process. When the process was applied to a power plant, it led to an 80% increase in the cost of electricity (Rao and Rubin, 2002). Unlike the costly aqueous amine-based capture, CO₂ capture using amine-impregnated solid sorbent is attracting researchers' attention with the potential to reduce energy consumption, improve regeneration, enable greater capacity, offer selectivity, and provide ease of handling (Samanta et al., 2012). The calcium looping process is another post-combustion technology, which exploits the reversible gas–solid reaction between calcium oxide (CaO) and CO₂ to produce calcium carbonate (CaCO₃)

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Nomenclature

A	amplitude of reaction
Am_0	initial concentration of amine in sorbent
C	mass concentration of species (mol m^{-3})
c_p	specific heat ($\text{J kg}^{-1} \text{K}^{-1}$)
D	diffusion coefficient ($\text{m}^2 \text{s}^{-1}$)
d_{pore}	pore size (cm)
d_p	diameter of particle (m)
ΔH	enthalpy of reaction (J mol^{-1})
ΔS	enthalpy of reaction ($\text{J mol}^{-1} \text{K}^{-1}$)
E	activation energy of reaction (J mol^{-1})
G	generation/consumption rate ($\text{mol m}^{-3} \text{s}^{-1}$)
K	equilibrium constant of reaction
R	universal gas constant ($8.314 \text{ J mol}^{-1} \text{K}^{-1}$)
R	reaction rate ($\text{mol m}^{-3} \text{s}^{-1}$)
r	radial direction (m)
h_i	convection mass transfer coefficient of species i (m s^{-1})
h_T	convection heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
i	species index
M	molecular weight (kg mol^{-1})
Nu	Nusselt number
P	ambient pressure (Pa)
Pr	Prandtl number
Re	Reynolds number
Sc	Schmidt number
Sh	Sherwood number
T	Kelvin temperature (K)
t	time (s)

Superscripts and subscripts

d	dry adsorption of CO_2
i	species
j	reaction
k	space index of particle
p	water adsorption
S	on outer surface
w	wet adsorption of CO_2

Greek

α	coefficient of finite difference equation in mass transfer
β	coefficient of finite difference equation in heat transfer
ε	porosity
κ	conductivity ($\text{W m}^{-1} \text{K}^{-1}$)
ρ	density (kg m^{-3})
μ	viscosity (Ns m^{-2})
σ	conductivity (S m^{-1})
ν	stoichiometry

(Abanades et al., 2004). The technology also shows a good prospect for capturing CO_2 effectively (Manovic and Anthony, 2010; Broda and Muller, 2012).

Modeling and simulation studies have been performed on several carbon capture systems for improved performance at different scales. For example, the CO_2 absorption/stripping process with aqueous monoethanolamine (MEA) was modeled based on thermodynamics and reaction rates (Freguia and Rochelle, 2003). The researchers analyzed process variables to find operating conditions at a low-steam requirement. To find the best configuration of the CO_2 stripper, the performance of several alternative designs was simulated and compared (Oyenekan and Rochelle, 2007). CO_2

absorption using a membrane contactor also was studied via modeling and experiment (Hoff et al., 2004). The absorption rate was calculated with changing temperature, liquid velocity, and CO_2 partial pressure. The two-dimensional diffusion-reaction model closely predicted the experimental results. A computational fluid dynamics (CFD)-based model for CO_2 stripping also was developed (Ghadiri et al., 2013). Using this model, the effect of liquid velocity and temperature on stripping performance was investigated. Studies using CFD-based simulation also examined the feasibility of converting existing coal-fired power plants to oxy-combustion (Tigges et al., 2009).

In addition, various modeling and simulation studies regarding CO_2 capture by solid sorbents have been performed. The CO_2 adsorption behavior of amine-functionalized mesoporous silica was modeled using chemical interaction with the amine group and physisorption on the adsorbent's surface (Serna-Gurerrero et al., 2010). The model was applied to several sorbents, and the CO_2 adsorption isotherm of each adsorbent was predicted. In zeolite imidazolate frameworks, the effect of functional group structure on CO_2 adsorption was investigated using molecular modeling (Morris et al., 2010). It showed that polarizability and structural symmetry were the key factors for high CO_2 uptake. Modeling was also used to evaluate the performance and cost of amine-based CO_2 capture systems (Rao and Rubin, 2006). The most cost-effective strategies and key factors in controlling CO_2 levels were proposed, with a fluidized bed reactor considered as the most promising design for CO_2 capture by solid sorbents (Yang and Hoffman, 2009). In this design, heat and mass transfer and chemical reactions in the reactor depend on the interactions of gas and solid phases in the bed (Smolders and Baeyens, 2001). To calculate the two-phase flow in the fluidized bed, several models have been developed (Kato and Wen, 1969; Kunii and Levenspiel, 1990; de Souza-Santos, 2007). When fluidized-bed hydrodynamics is combined with a CO_2 adsorption reaction, device-scale simulations as such can provide more detailed information for improving the reactor design. However, the CFD-based device-scale CO_2 capture predictions depend on accurate models in both the particle-scale reaction kinetics as well as in multi-phase flow simulations.

At the particle scale, CO_2 adsorption by an amine sorbent was simulated using a shrinking core model to elucidate the relative importance of pore diffusion and surface chemical reaction in controlling the rate of reaction (Monazam et al., 2011). Lee et al. (2011) proposed a phenomenological model for the adsorption kinetics of amine-impregnated mesoporous sorbent in the presence of water, and the model was fitted to experimental data obtained from thermogravimetric analyses. Recently, Xu et al. (2012) proposed a generalized kinetic model for heterogeneous gas–solid reactions with the introduction of a distinctive layer between reacted and not-yet-reacted regions.

In this study, adsorption phenomena are modeled on the particle scale based on three fundamental reactions occurring in the amine-impregnated mesoporous sorbent in a CO_2 rich environment. The model calculates on the particle scale (1) how different adsorption reactions progress; (2) how the reactions depend on each other; and (3) the amount of each species adsorbed on the sorbent particle resulting from the reactions. Similar particle-based models have been developed and used in several other research areas, i.e., metal sorption with resins (Juang and Lin, 1995), olefin polymerization (McKenna and Soares, 2001), and ozone uptake (Smith et al., 2002), but only for single reaction. In the current study, three different reactions occur simultaneously in the amine-based sorbent particle, assisting each other in one way and hindering in another, which makes particle-scale CO_2 adsorption prediction more difficult. With the successfully developed particle scale reaction kinetics model and the numerical solution algorithm presented here, mass transfer and adsorption phenomena resulted via different reaction

Table 1
Three reactions in sorbent particles.

Dry adsorption	$2R_2NH + CO_2 \rightleftharpoons R_2NH_2^+ + R_2NCO_2^-$ (3)
Wet adsorption	$R_2NH + CO_2 + H_2O(phy) \rightleftharpoons R_2NH_2^+ + HCO_3^-$ (4)
Physisorption of water	$H_2O(g) \rightleftharpoons H_2O(phy)$ (5)

mechanisms are quantitatively predicted at different spatial locations within the particle for quantities that are difficult to measure experimentally (Monazam et al., 2011; Lee et al., 2011), enabling further understanding and improvement of amine-based sorbent particle design.

2. Governing equations

In this study, carbon dioxide adsorption/desorption is considered in a spherically-shaped sorbent particle. The sorbent particle is made from mesoporous silica incorporated with amine (Fauth et al., 2012). When the particle is placed in a CO₂-rich environment, CO₂ diffuses into the tiny pores of the sorbent particle. During the diffusion process, some CO₂ molecules are incorporated with amine by adsorption, and other CO₂ molecules keep moving inside the particle. Eventually, adsorption of CO₂ by the sorbent is saturated. Mass transfer in a spherical sorbent particle is described by the following equation:

$$\frac{\partial C_i}{\partial t} = D_i \cdot \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial C_i}{\partial r} \right) + G_i, \quad i = [1], [2]. \quad (1)$$

C_i is the concentration of a species (i), D_i is its respective diffusivity, and G_i represents the generation or consumption rate of the species in a sorbent particle. Water vapor, which is involved in the wet adsorption reaction, has the same form of mass transfer equation as Eq. (1). The pore size within sorbent particle has a nanoscale length, and Knudsen diffusion is the dominant type of mass transfer. Diffusivity (in cm² s⁻¹) is estimated from Knudsen diffusivity:

$$D_i = \varepsilon \cdot (4850 d_{\text{pore}} \sqrt{T/M_i}), \quad (2)$$

where ε is porosity, d_{pore} is pore diameter in cm, T is temperature in Kelvin, and M_i denotes molecular mass of species i (g mol⁻¹). Table 1 lists the three reactions considered in this model. The index (i) is mapped to different species in Table 1. For example, [1] is the index for CO₂, and [2] is for water vapor.

In the table, R₂NH is ethylenediamine supported on silica, a sorbent material. CO₂ adsorption by amine is realized through complicated steps. This process is well explained in the references (Vaidya and Kenig, 2007; Choi et al., 2009; Sanz et al., 2010). For the purpose of this model, CO₂ is assumed to be adsorbed via two reactions: dry and wet adsorption. When there is no moisture around the sorbent, CO₂ reacts with amine as in Eq. (3) of Table 1. However, when water is present, another reaction is available to CO₂ adsorption as seen in Eq. (5) (Satyapal et al., 2001). NETL recently developed an amine-impregnated mesoporous sorbent, NETL-32D, and reported its thermodynamic and kinetic data via thermal gravimetric analysis (TGA) (Lee et al., 2011). This study uses their data to calculate these adsorption rates. The rate of dry reaction,

Table 3
Model parameters.

Parameter	Symbol	Value
Porosity of sorbent particle	ε	0.1%
Pore size (cm)	d_{pore}	338E-8
Particle diameter (μm)	d_p	150
Initial amine concentration	$Am0$	2531
Density of particle (kg m ⁻³)	ρ	442
Specific heat (J kg ⁻¹ K ⁻¹)	c_p	1130
Conductivity (W m ⁻¹ K ⁻¹)	κ	1
Operating pressure (Pa)	P	1.0E5

identified by subscript d, is defined by the generation rate of carbamate (R₂NCO₂⁻):

$$R_d = \frac{\partial C_5}{\partial t} = \frac{A_d \cdot T \cdot P}{Am0} \cdot \exp \left(-\frac{E_d}{R \cdot T} \right) \cdot \left(\frac{C_1 \cdot C_3^2}{R \cdot T \cdot P} - \frac{C_4 \cdot C_5}{K_d} \right) \quad (6)$$

where P is the ambient pressure, $Am0$ is the initial concentration of amine in the sorbent, and C_i is the concentration of each species involved in the reaction. A_d and E_d are the dry reaction amplitude and activation energy, respectively. K_d is the equilibrium constant defined in the following:

$$K_d = \exp \left(\frac{\Delta H_d - T \cdot \Delta S_d}{R \cdot T} \right), \quad (7)$$

where ΔH_d and ΔS_d are the enthalpy and entropy of the dry reaction, respectively.

The wet reaction rate, identified by subscript w, is equal to the generation rate of bicarbonate (HCO₃⁻):

$$R_w = \frac{\partial C_7}{\partial t} = \frac{A_w \cdot T \cdot P}{Am0} \cdot \exp \left(-\frac{E_w}{R \cdot T} \right) \cdot \left(\frac{C_1 \cdot C_3 \cdot C_6}{R \cdot T \cdot P} - \frac{C_4 \cdot C_6}{K_w} \right) \quad (8)$$

For the water adsorption reaction, identified by subscript p:

$$R_p = \frac{\partial C_6}{\partial t} = A_p \cdot T \cdot P \cdot \exp \left(-\frac{E_p}{R \cdot T} \right) \cdot \left(\frac{C_2}{R \cdot T \cdot P} - \frac{C_6}{K_p} \right) \quad (9)$$

The equilibrium constants, K_w and K_p , are defined the same as K_d . Table 2 illustrates the kinetic and thermodynamic values of each reaction. The species produced from the reactions in Table 1, i.e., $i = [3] \sim [7]$, are assumed not to diffuse in the particle, and their mass transfer is reduced to Eq. (10):

$$\frac{\partial C_i}{\partial t} = G_i \quad i = [3], [4], \dots, [7] \quad (10)$$

The source term of a species (i) for Eq. (1) or (10) is calculated as follows:

$$G_i = \sum_j v_{ij} \cdot R_j, \quad (11)$$

v_{ij} is the stoichiometry of species i in the reaction j of Table 2.

In the sorbent particle, the heat transfer equation takes a form similar to the mass transfer equation. Heat generation is calculated with reaction rate and enthalpy:

$$\rho \cdot c_p \frac{\partial T}{\partial t} = \kappa \cdot \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + L, \quad L = \sum_j R_j \cdot \Delta H_j, \quad (12)$$

where ρ , c_p , and κ are explained in Table 3. Their values also are shown in the table.

Table 2
Kinetic and thermodynamic properties of reactions in a sorbent particle.

Reaction (j)	A_j	E_j (J mol ⁻¹)	ΔH_j (J mol ⁻¹)	ΔS_j (J mol ⁻¹ K ⁻¹)
Dry reaction (d)	10 ²	57,700	-64,700	-174.6
Wet reaction (w)	10 ^{0.418}	58,200	-36,300	-88.1
Water physisorption (p)	10 ^{2.75}	28,200	-52,100	-78.5

Mass and heat transfer between the particle and its surrounding gas takes place at the sorbent particle's outer surface and provided boundary conditions for Eqs. (1) and (12):

$$D_i \frac{\partial C_i}{\partial r} = h_i(C_{i|^\infty} - C_{i|S}), \quad i = [1], [2] \quad \text{on } r = S \quad (13)$$

$$k \frac{\partial T_i}{\partial r} = h_T(T|^\infty - T|S), \quad \text{on } r = S. \quad (14)$$

$C_{i|^\infty}$ and $C_{i|S}$ are the concentration of species i in the surrounding gas and on the particle's surface. h_i/h_T are the convection mass/heat transfer coefficients, which are estimated using the following empirical relations (Ruthven, 1984):

$$Sh_i = \frac{d_p h_i}{D_i} = 2.0 + 1.1 Sc^{1/3} Re^{0.6} \quad (15)$$

$$Nu = \frac{d_p h_T}{k} = 2.0 + 1.1 Pr^{1/3} Re^{0.6}. \quad (16)$$

Wakao and Funazkri (1978) analyzed many experimental data of particle-fluid mass transfer in packed bed systems and proposed Eq. (15). When the velocity of a gas flowing through the bed becomes small, the Sherwood number of a particle goes to 2.0. The Prandtl number (Pr) of the surrounding gas is set as 0.7, which is the Pr of air at 350 °C (Bergman et al., 2011). The Re (Reynolds number) and Sc (Schmidt number) are calculated with 20E–6 m² s^{−1} of gas viscosity and 0.5 m s^{−1} of average gas velocity. d_p is the diameter of the sorbent particle.

At the center of the particle, i.e., $r=0$, another boundary condition is given by symmetry constraint:

$$\frac{\partial C_i}{\partial r} = 0, \quad (17)$$

$$\frac{\partial T}{\partial r} = 0. \quad (18)$$

The diffusion term in Eqs. (1) and (12) can be further simplified at $r=0$ when it is discretized into a finite difference form according to Gutierrez-Miravete (2006):

$$\lim_{r \rightarrow 0} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) = 3 \left(\frac{\partial^2 T}{\partial r^2} \right). \quad (19)$$

3. Solution algorithm

The finite difference method is used to solve heat and mass transfer equations (Mitchell and Griffiths, 1980; Suh and Park, 2012). Eq. (1) is converted into a finite difference approximation form based on the Crank–Nicolson scheme:

$$\begin{aligned} & -\alpha_{k+1} C_{i,k+1}^{t+\Delta t} + (2 + \alpha_{k+1} + \alpha_{i-1}) C_{i,k}^{t+\Delta t} - \alpha_{k+1} C_{i,k-1}^{t+\Delta t} \\ & - G_i(C_{[1]_k}^{t+\Delta t}, \dots, C_{[7]_k}^{t+\Delta t}) = \alpha_{k+1} C_{i,k+1}^t + (2 - \alpha_{k+1} - \alpha_{k-1}) C_{i,k}^t \\ & + \alpha_{k+1} C_{i,k-1}^t + G_i(C_{[1]_k}^t, \dots, C_{[7]_k}^t), \quad \text{for } i = [1], [2] \end{aligned} \quad (20)$$

where k and t are the indices for space and time, respectively. Δt is a time step, while α is the coefficient of the approximation equation. G_i , generation/consumption of C_i , is presented as a function of the concentration of species involved in the reaction. For the concentration of the species other than CO₂ and water vapor, Eq. (21), or the approximation form of Eq. (12), is used:

$$\begin{aligned} & C_{i,k}^{t+\Delta t} - \frac{\Delta t}{2} G_i(C_{[1]_k}^{t+\Delta t}, C_{[1]_k}^{t+\Delta t}, \dots, C_{[7]_k}^{t+\Delta t}) = C_{i,k}^t \\ & + \frac{\Delta t}{2} G_i(C_{[1]_k}^t, \dots, C_{[7]_k}^t) \quad \text{for } i = [3] \sim [7]. \end{aligned} \quad (21)$$

With known $C_{i,k}^t$ values, the concentrations of each species after Δt are calculated by iteration. $C_{i,k}^{t+\Delta t}$ is iteratively updated until the prescribed convergence criterion is met.

We first attempted at calculating the species generation terms on the left sides of Eqs. (20) and (21) with $C_{i,k}^{t+\Delta t}$, i.e., the concentrations calculated from the previous iteration step. However, this approach leads to solution fluctuations in some cases. As a numerically more robust approach, the generation term is changed to a function of the current value of $C_{i,k}^{t+\Delta t}$, and all of the concentrations are calculated by solving the nonlinear system of equations of Eqs. (20) and (21). One more relationship/constraint is satisfied during computation:

$$C_{[3]_k}^t + C_{[4]_k}^t + C_{[5]_k}^t = \text{constant} = Am_0 \quad \text{for all } k, \text{ and } t. \quad (22)$$

The total concentration of amine ([3]), protonated amine ([4]), and carbamate ([5]) are kept constant for all positions and times because species [4] and [5] are produced from species [3]. The constant value is given as one input value.

The heat transfer equation, Eq. (12), is discretized in a similar form as in Eq. (20):

$$\begin{aligned} & -\beta_{k+1} T_{k+1}^{t+\Delta t} + (2 + \beta_{k+1} + \beta_{t-1}) T_k^{t+\Delta t} - \beta_{k+1} T_{k-1}^{t+\Delta t} - L_k^{t+\Delta t} \\ & = \beta_{k+1} T_{k+1}^t + (2 - \beta_{k+1} - \beta_{k-1}) T_k^t + \beta_{k+1} T_{k-1}^t + L_k^t \end{aligned} \quad (23)$$

In this equation, β s are coefficients of the approximation equation. The amount of heat generation (L) has been already determined by each adsorption reaction rate.

The overall numerical solution algorithm is illustrated in the flowchart shown in Fig. 1. As a computational tool, Mathematica® V9.0 is used to program both the solving process for differential equations and the post-process of visualization results.

4. Results and discussion

When CO₂ gas flows around a sorbent particle in the absence of water, the gas combines with an amine (R₂NH) through a dry adsorption reaction. The reaction starts from the particle's outer shell and proceeds toward the interior. Eventually, CO₂ can be captured over the whole particle. When 7.5 mol% CO₂–0 mol% H₂O gas is supplied to a sorbent particle at a temperature of 108 °C, the evolutions of reaction rate and species concentration along the particle radius are calculated as shown in Fig. 2. The reaction starts at a high rate on the particle surface, see Fig. 2(a). As the reaction front moves inside of the particle, its peak value decreases. However, the width of the reaction zone is increased. As the front approaches the center, the peak increases somewhat. After the reaction zone is extended over all radial positions, its rate quickly decreases. Then, the reaction reaches equilibrium. When adsorption changes with time as in Fig. 2(a), the concentrations of the amine and carbamate ion (R₂NHCOO[−]) are calculated as shown in Fig. 2(b) and (c).

To determine the influence of grid size on the simulation results, we carried out the preceding calculations with three different numbers of nodes along the particle radial direction: 10, 20, and 30. The results indicate that dry adsorption rate changes more dramatically than the other output variables, i.e., concentration and temperature. The dry adsorption rate distributions of the three models are compared in Fig. 3. For times less than 30 s, significantly different dry adsorption rates are predicted by the three models with different grid resolutions. Compared with the other two models, the 10 node model predicts much lower rate results. However, at 30 s, the distributions become smoother, and the difference between different grid resolutions is quite small.

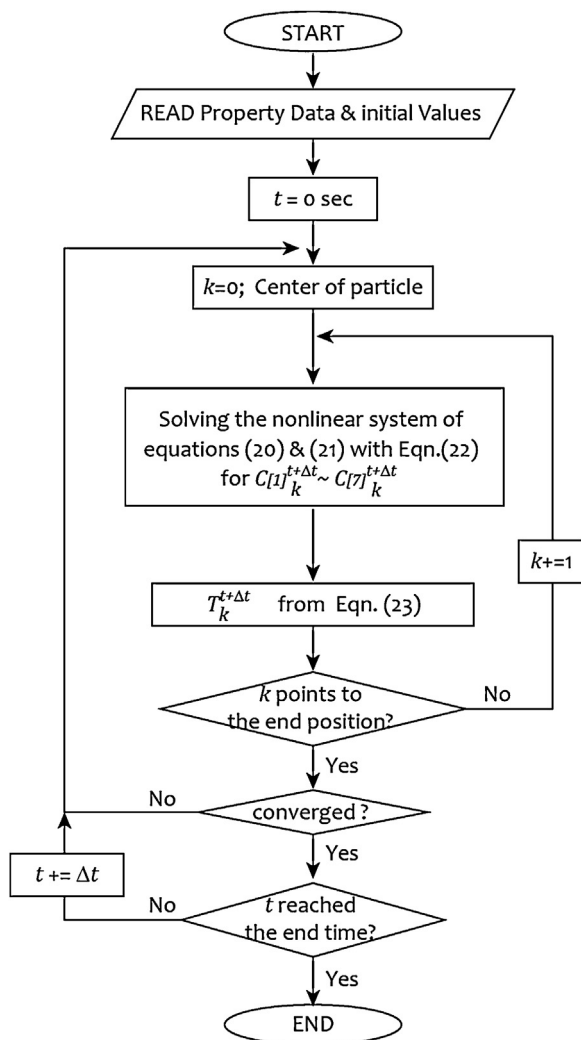


Fig. 1. Flow chart of numerical solving process.

The volume average value ($\bar{A}$) over a sorbent particle is defined as the following equation:

$$\bar{A} = \frac{\int A(r) \cdot r^2 dr}{\int r^2 dr}. \quad (24)$$

The average adsorption rate of the particle is calculated with the reaction time as shown in Fig. 4(a). After starting at a high rate, the reaction slows down. However, as the reaction gets closer to the center of the particle, the average rate increases. After achieving the peak rate at 25 s, the reaction ceases and finally reaches a state of equilibrium throughout the particle. Grid size sensitivity is reduced in the average adsorption rate (shown in Fig. 4(a)). The average concentration profiles of carbamate shown in Fig. 4(b) are almost the same for the three different grid sizes. For computational efficiency, the 20-node model is adopted as the standard grid size in this study, and the simulation results presented below are all based on this model.

4.1. Comparison of simulation with experiment

Lee et al. (2011) used TGA to study reaction kinetics and equilibriums of the sorbent materials. When several combinations of CO₂ gas and vapor water passed through sorbent particles, Lee et al. measured the weight change of the sample with changing temperature in the chamber. They estimated thermodynamic parameters

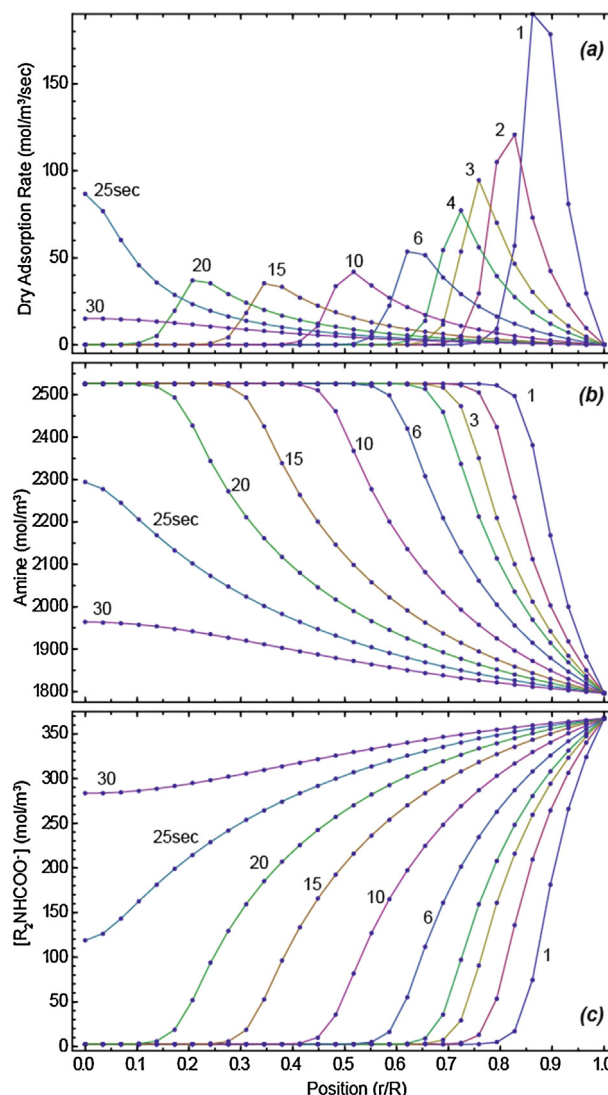


Fig. 2. Evolution of 7.5 mol% CO₂–0 mol% H₂O gas adsorption in a sorbent particle at 108 °C. (a) reaction rate, (b) amine, and (c) carbamate.

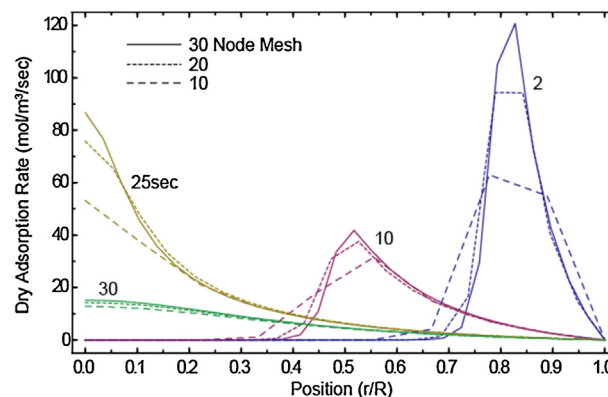


Fig. 3. Computing adsorption rate in a sorbent particle with three different grid sizes.

of reactions in sorbent particles as shown in Table 2. To validate the model developed in this study, adsorption in a sorbent particle is simulated under the conditions employed by Lee et al. (2011), and the predicted results are compared with their experimental data.

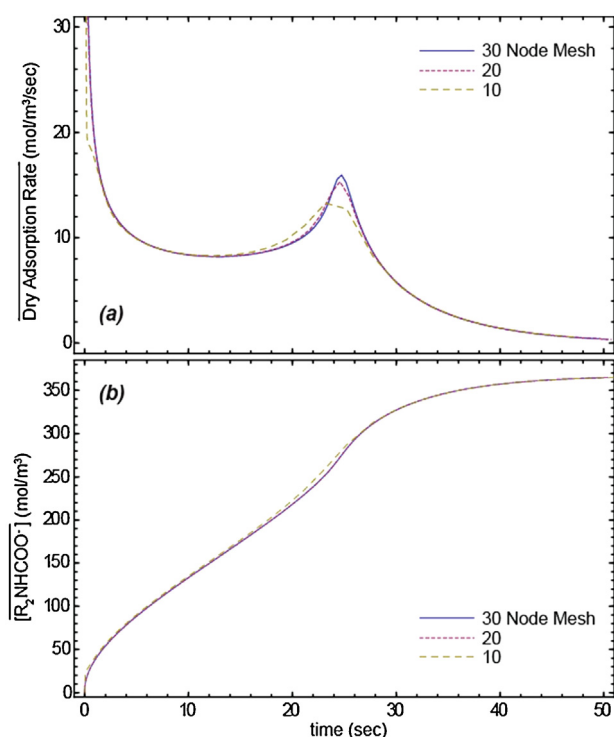


Fig. 4. Evolution of average (a) adsorption rate and (b) carbamate produced in a sorbent particle.

In the experiment, the chamber temperature was sequentially set to 108 °C, 100 °C, 92 °C, 65 °C, 57 °C, 49 °C, 41 °C, 65 °C, and 108 °C. Each step was held for 90 min. To simulate the carbon capturing process of a sorbent particle, this temperature history is applied to $T|^\infty$ in Eq. (13). Under the condition of 7.5 mol% CO₂–0% H₂O, the particle's average temperature ($\bar{T}$) is calculated in the dotted curve of Fig. 5, which is similar to the ambient temperature, $T|^\infty$. Fig. 5 also shows the chamber temperature measured during the experiment. The actual temperature fluctuates, and the transitions to the next temperature are delayed with respect to the condition stated in the literature (Lee et al., 2011), which may contribute to the differences between the simulation results and the experimental measurements.

In the absence of water around the sorbent, the particle gains weight only through the dry adsorption reaction, and its average weight gain ($\overline{WG}$) is calculated as in Fig. 6. As the chamber temperature decreases, the exothermic reaction intensifies, and more

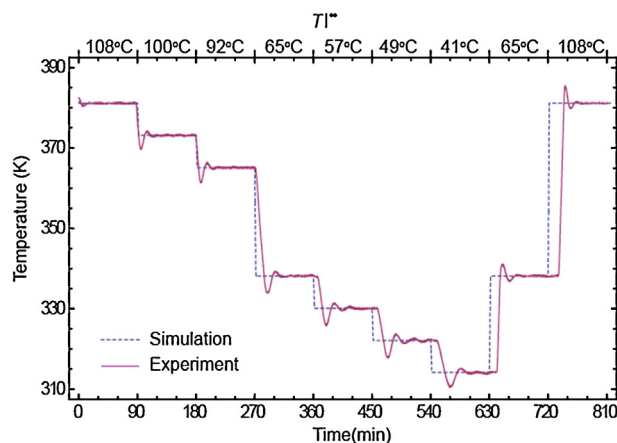


Fig. 5. Temperature change of sorbent (simulation) and chamber (experiment).

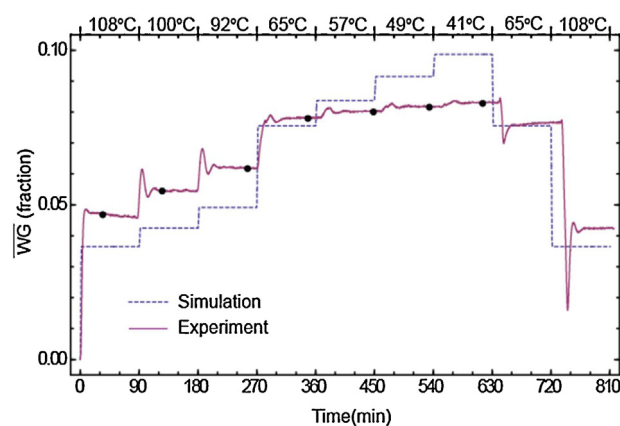


Fig. 6. Change of average weight gain of sorbent particle at 7.5 mol% CO₂–0 mol% H₂O gas.

CO₂ gases are captured in it. This trend is observed in both simulation and experiment curves, but there is a difference between their absolute values. To check the simulation's accuracy, weight gain is also calculated using thermodynamic equilibrium.

Equilibrium concentrations of a dry reaction can be estimated using only thermodynamic parameters. First, the site fraction of carbamate ($R_2NCO_2^-$) is assume to be zero initially, and changed to x at equilibrium condition. The equilibrium constant (K_d) of dry reaction is related to x and thermodynamic parameters (Lee et al., 2011):

$$K_d = \frac{x \cdot x}{P_{CO_2} \cdot (1 - 2x)^2} = \frac{\exp(-(\Delta H_d - T \cdot \Delta S_d)/R \cdot T)}{P} \quad (25)$$

With ΔH_d and ΔS_d in Table 2, the equilibrium concentration (x) of $R_2NH_2^+$ or $R_2NCO_2^-$ and the resulting weight gains are calculated for each temperature. The calculated values in the third and sixth columns of Table 4 are equal to the values from the simulation. However, compared to the experimental values, they are smaller at high temperatures and larger at low temperatures. A similar comparison is made for the water adsorption reaction (9 mol% H₂O only), and the gap between them is determined (shown in both Table 4 and Fig. 8). These discrepancies indicate that the thermodynamic parameters extracted by Lee et al. (2011) did not accurately represent the experimental results. Lee et al. (2011) did report some anomalies with their experiments and explained the difficulties in accurately determining the parameters.

In this study, mass and heat transfer coefficients between sorbent particles and gas are estimated using empirical equations, Eqs. (15) and (16), proposed by Wakao and Funazkri (1978). The coefficients are functions of the gas flow velocity in the packed bed, that is, the Reynolds number. To check the sensitivity of the coefficients on simulation, weight gain is calculated under two different gas velocities in the bed: 0.5 m s⁻¹ and near 0.0 m s⁻¹. Fig. 7 shows weight gain change during the initial 3 minutes under Fig. 6 condition. Even when mass and heat transfer coefficients are reduced to low-limit values, no difference is seen between the two weight gain curves. This means the rate of weight gain mostly depends on the diffusion and reaction inside the sorbent particle, and the discrepancy between experimental measurements and simulation results is unrelated to the uncertainty in mass and heat transfer coefficients. However, the concentration of CO₂ around the particles could be depleted from the top surface to inside the particle bed. When CO₂ adsorption becomes enhanced, the gradient of CO₂ concentration increases along the bed height and CO₂ cannot reach below particular position in the bed. This offers a reason for the difference between experiment and simulation as simulation assumes a fixed boundary concentration value. As shown in

Table 4
Comparison of equilibrium weight gains from calculation and measurement.

T (°C)	x (mol%)	Weight gain, $\frac{M_{\text{CO}_2} \cdot A_{\text{m}0} \cdot x}{\rho}$	Measurement (●) in Fig. 6	y (mol)	Weight gain, $\frac{M_{\text{H}_2\text{O}} \cdot y}{\rho}$	Measurement (●) in Fig. 7
108	0.145	0.099	0.08318	98.6	0.004	0.00075
100	0.169	0.092	0.08186	140.3	0.006	0.00145
92	0.195	0.084	0.08026	202.7	0.008	0.00306
65	0.300	0.076	0.07822	797.8	0.033	0.02238
57	0.332	0.049	0.06202	1250.2	0.051	0.04398
49	0.363	0.043	0.05443	2003.0	0.082	0.08360
41	0.392	0.037	0.04655	3287.1	0.134	0.09610
Dry adsorption (7.5 mol% CO ₂ only)				Water adsorption (9.0 mol% H ₂ O only)		

Fig. 6, the weight gain of experiment becomes limited as temperature decreases below 65 °C, but the simulation value continues to rise above the experimental value. Assuming that CO₂ can be supplied down to the bottom of the particle bed at 65 °C, where the difference between experiment and simulation is at the minimum, a CO₂ depletion zone increases as temperature decreases further than 65 °C. About 17% of the sorbent bed from its bottom does not contribute to weight gain via CO₂ adsorption at 41 °C.

Nevertheless, with the set of thermodynamic parameters proposed by Lee et al. (2011), our simulation results are able to capture the overall temperature-dependent adsorption behaviors as those observed in experiments. Hence the work can provide a better understanding of CO₂ adsorption phenomena occurring in a sorbent particle and help further improve solid sorbent-based CO₂ sequestration system.

4.2. Wet adsorption

The CO₂ adsorption mechanism is known to be dependent on the presence of water (Kohl and Nielsen, 1997; Satyapal et al., 2001). In the absence of water, it takes two moles of amine to adsorb one mole of CO₂ (refer to Eq. (3)). However, under wet conditions, one mole of amine can adsorb one mole of CO₂. As such, the expectation is that a sorbent's ability to capture CO₂ is enhanced in the presence of water. Fig. 9 shows the predicted sorbent's weight gain under 7.5 mol% CO₂–9 mol% H₂O condition, along with experimental data in the literature (Lee et al., 2011). Compared with Fig. 6 (dry condition), the sorbent gains more weight under wet condition in both simulation and experiment. Nonetheless, that incremental weight gain might not imply an increase in captured CO₂. The sorbent could gain weight by adsorbed H₂O. To clarify this point, total weight gain is divided into each species' contribution.

In Fig. 10, the lower pink region is the CO₂ contribution to the weight gain, and the upper blue region is the H₂O

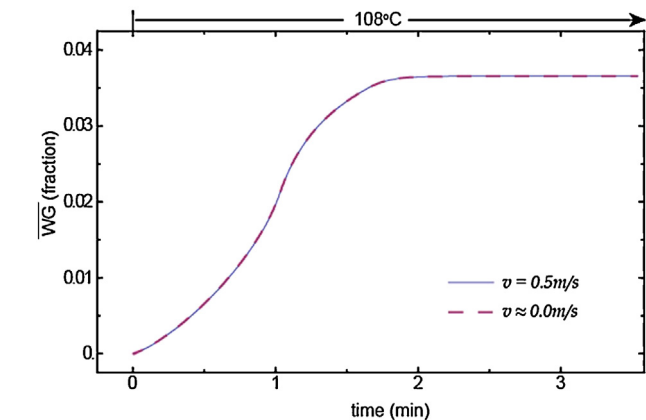


Fig. 7. Change of average weight gain of sorbent particle during initial 3 min in Fig. 6 at two different gas velocity conditions.

contribution to the weight gain. In the first time period, with temperature remaining at 108 °C, CO₂ adsorption accounts for more than 85% of the weight gain. As the temperature decreases to 41 °C, the weight gain of sorbent increases, but the CO₂ portion in it becomes less than 50% because H₂O adsorption is enhanced much more than CO₂ adsorption at the lower temperatures. Water is physically adsorbed into the sorbent particle (Eq. (5)), but some of physisorbed water, H₂O(phy), is consumed by a wet reaction (Eq. (4)) and turns into other adsorbates. To compensate for reduction in H₂O(phy), the physisorption reaction adsorbs water further and generates H₂O(phy), which again promotes the wet adsorption reaction. The two reactions, Eqs. (4) and (5), interact with each other in this way and eventually reach equilibrium states of their own.

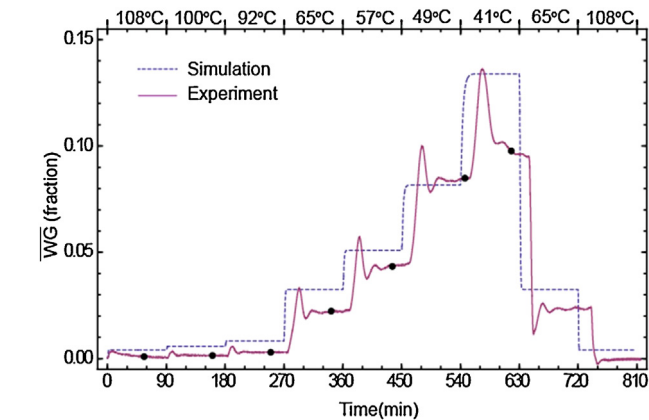


Fig. 8. Change of average weight gain of sorbent particle at 0 mol% CO₂–9 mol% H₂O gas.

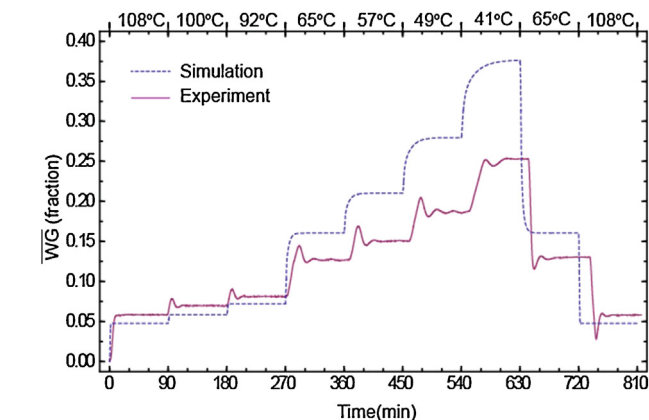


Fig. 9. Change of average weight gain of sorbent particle at 7.5 mol% CO₂–9 mol% H₂O gas.

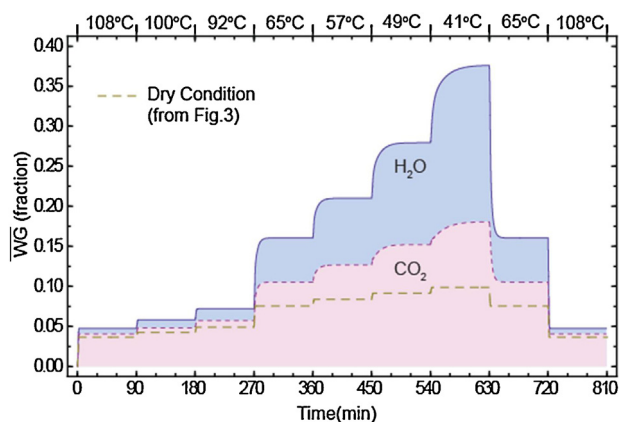


Fig. 10. Change of average weight gains caused by CO_2 and H_2O in sorbent particle at 7.5 mol% CO_2 –9 mol% H_2O gas.

The weight gain from water (Fig. 10) is divided into two forms shown in Fig. 11. In the two forms, physisorbed H_2O and H_2O in the other adsorbates, water is present in the sorbent under all temperature conditions. In the latter form, the amount increases along with the wet reaction at lower temperatures. The weight gain of H_2O that remains in the physisorbed form is almost equal to the weight gain in Fig. 8 because equilibrium H_2O concentration is the same in both conditions. In other research (Satyapal et al., 2001; Lee et al., 2011), the amount of CO_2 captured under wet conditions was estimated. However, the estimated value could be overestimated because the amount of H_2O adsorbed was assumed to be the same as that under H_2O -only conditions. According to Fig. 11, H_2O in the other form is responsible for a 6% weight gain at 41 °C, which is about 35% of the weight gain by CO_2 at the same temperature. In Fig. 10, the weight gain by CO_2 is not only calculated under wet conditions (9 mol% H_2O), but also under dry conditions. When the two dotted curves of that figure are compared, one can easily quantify how much more favorable the wet condition is for CO_2 capture. The difference between the two dotted curves grows with decreasing temperature, reaching maximum at 41 °C, where CO_2 is adsorbed more in the wet condition than in the dry condition by 0.082, which amounts to 82% of the weight gain via dry adsorption. Thus, it is evident that the wet condition has the advantage over the dry condition in capturing CO_2 , and the wet effect on CO_2 adsorption becomes more significant at lower temperatures. It was estimated that three times more CO_2 would be removed in the presence of water than in the absence of water (Satyapal et al., 2001). Still, without considering H_2O in the other adsorbates, they

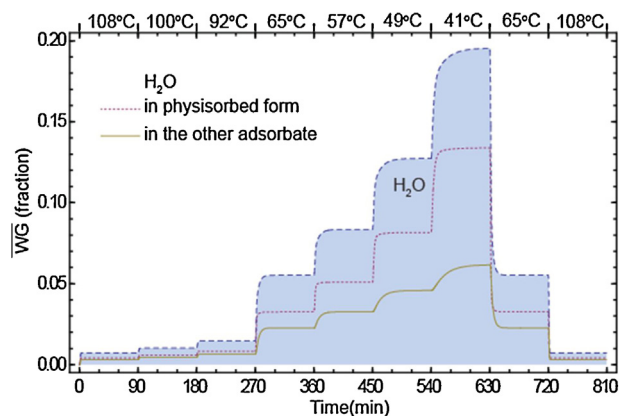


Fig. 11. Change of average H_2O weight gains in physisorbed form and the other adsorbate at 7.5 mol% CO_2 –9 mol% H_2O gas.

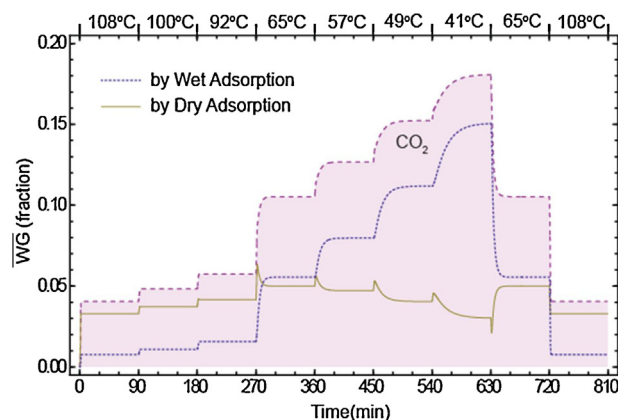


Fig. 12. Change of CO_2 weight gains via wet and dry adsorptions at 7.5 mol% CO_2 –9 mol% H_2O gas.

overestimated the amount of adsorbed CO_2 . Fig. 10 shows that CO_2 is captured 1.75 times more in the wet condition even at the lowest temperature, 41 °C.

To analyze the CO_2 adsorption path, CO_2 weight gains via wet and dry adsorptions are calculated separately (Fig. 12). At temperatures higher than 92 °C, dry adsorption is more dominant in capturing CO_2 than wet adsorption. However, the effect of the two reactions reverses at 65 °C. As the reaction temperature continues to decrease, CO_2 capture via wet adsorption keeps increasing, but the effect of dry adsorption decreases. When the Gibbs free energy changes of the two reactions are compared using the data in Table 2, wet adsorption is more favorable at all reaction temperatures in Fig. 12. But, the wet reaction is severely limited by H_2O physisorbed in the sorbent at higher temperatures. The physisorbed H_2O concentration is so low that the equilibrium concentration of adsorbed CO_2 is lower in the wet reaction than in the dry reaction. As the temperature decreases, the concentration of physisorbed H_2O increases enough for wet adsorption to exceed dry adsorption at 65 °C. At even lower temperatures, the wet adsorption effect expands further, and the equilibrium concentration of adsorbed CO_2 declines in the dry reaction because an amine (R_2NH) site for the dry reaction is reduced.

As shown in Fig. 12, the time for CO_2 adsorption to reach equilibrium changes with reaction temperature. The rate of adsorption reactions, given in Eqs. (6), (8), and (9), depends on the temperature, and it slows down at lower temperatures. This also is observed in the experiment depicted in Fig. 9, which shows it takes more time for the measurement to stabilize at low temperatures than at high temperatures.

Based on these temperature effects, it can be inferred that more CO_2 molecules can be separated from flue gas at lower adsorption temperature, but it needs more time for full adsorption. When dry adsorption time is compared with that of wet adsorption, dry is faster than wet (as shown in the kinetic parameter of Table 2, as well as Figs. 8 and 9). However, when the two adsorptions occur simultaneously in the presence of water, the stabilizing time for dry adsorption increases to that of wet adsorption as shown in Fig. 12. This is due to the two adsorptions' dependence on each other, and wet adsorption—the slower of the two—controls the overall reaction rate.

5. Conclusions

To analyze CO_2 adsorption behavior of an amine-based sorbent particle, a mathematical model is developed and solved numerically using the finite difference method under several reaction conditions, including at several temperatures

and with/without water. The simulation results demonstrate quantitatively how CO₂ is captured in the particle via different reaction mechanisms. To validate the model and the numerical solution procedures, simulation results are compared with the experimental data. The developed model reasonably simulates adsorption behaviors. The amount of each adsorbed species is calculated under changing conditions, which is hard to measure experimentally. The simulation reveals that the presence of water enhances CO₂ adsorption at 41 °C by 1.75 times, which is less than expected in other literatures. This is due to the weight gain by the adsorbed H₂O, which increases much more in the presence of water than under dry condition. As temperature decreases, both dry and wet adsorptions of CO₂ become more favorable. However, wet adsorption with a higher thermodynamic driving force intensifies so much that dry adsorption actually declines. At lower temperatures, dry adsorption of CO₂ and physisorption of H₂O become slower and are limited by the wet adsorption rate.

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